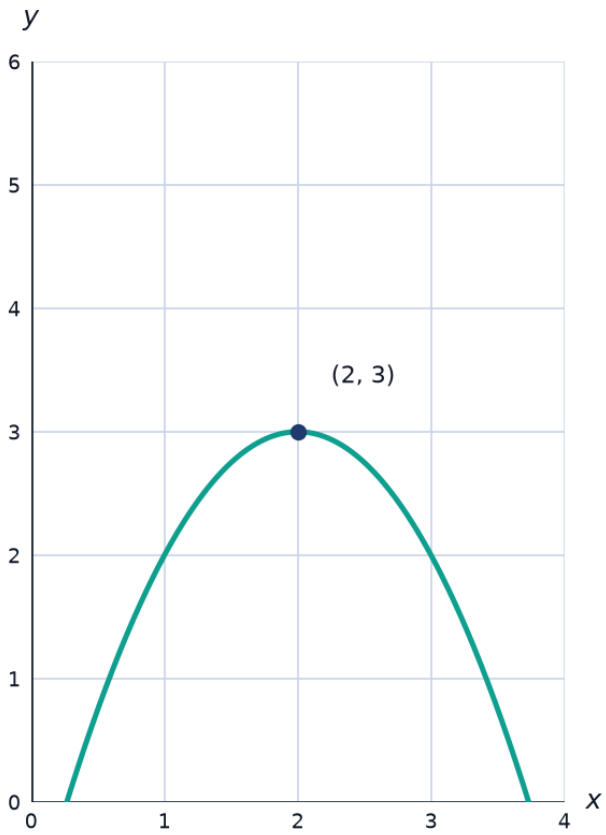
Implicit Differentiation and Inverse Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

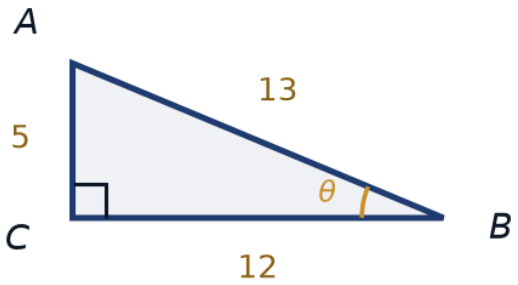
1 [Foundational] • Target: First Derivative by Implicit Differentiation

The graph of a differentiable function $y = f(x)$ is shown below. The tangent line to the graph at $x = 2$ has the equation $y = -x + k$. Find the value of k.



2 [Foundational] • Target: Second Derivatives Implicitly

Consider the function $y = g(x)$ and its inverse $x = h(y)$. The tangent to the graph of $g(x)$ at $x = a$ has a gradient of $\frac{1}{3}$. If $g(a) = 5$, what is the gradient of the tangent to the graph of $h(y)$ at $y = 5$? Show your complete algebraic work.



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3 [Foundational] • Target: Derivatives of Inverse Functions

Find $\frac{d}{dx}[\sin((-2x-2)^3)]$.

4 [Foundational] • Target: Derivatives of Inverse Trigonometric Functions

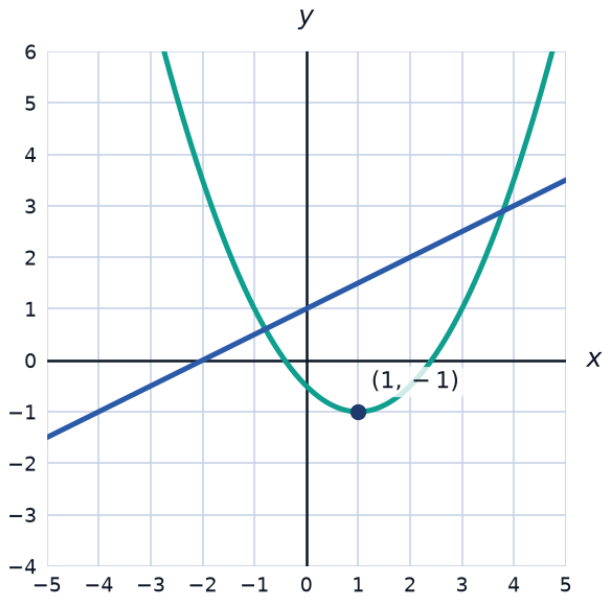
Find $\frac{d}{dx}[\sin((-2x-1)^3)]$.

5 [Foundational] • Target: First Derivative by Implicit Differentiation

Find $\frac{d}{dx}[(\sin(-2x)+-2)^3]$.

6 [Foundational] • Target: Second Derivatives Implicitly

Find $\frac{d}{dx}[(\sin(-2x)+-1)^3]$.



7 [Foundational] • Target: Derivatives of Inverse Functions

Find $\frac{d}{dx}[\sin((1-2x)^3)]$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Find $\frac{d}{dx}[\sin((-2x-2)^2)]$.

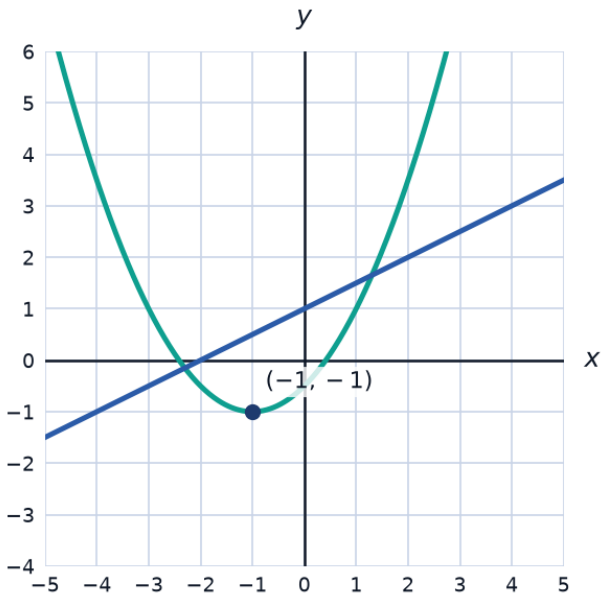
9 [Intermediate] • Target: First Derivative by Implicit Differentiation

Find $\frac{d}{dx}[\sin((-2x-1)^2)]$.

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10 **[Intermediate]** • **Target: Second Derivatives Implicitly**

Find $\frac{d}{dx}[\sin((3x+1)^3)]$.



11 **[Intermediate]** • **Target: Derivatives of Inverse Functions**

Find $\frac{d}{dx}[(\sin(-2x)+2)^3]$.

12 **[Intermediate]** • **Target: Derivatives of Inverse Trigonometric Functions**

Find $\frac{d}{dx}[\sin((3x-1)^3)]$.

13 **[Intermediate]** • **Target: First Derivative by Implicit Differentiation**

Find $\frac{d}{dx}[\sin((2x+2)^3)]$.

14 **[Intermediate]** • **Target: Second Derivatives Implicitly**

Find $\frac{d}{dx}[\sin((3x-2)^3)]$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

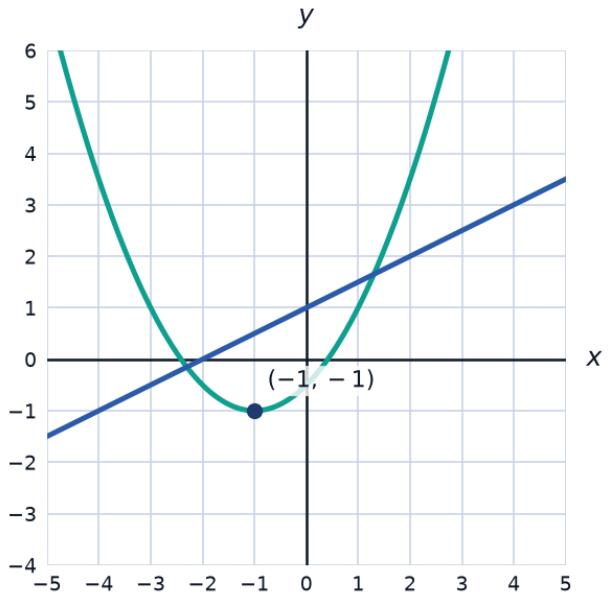
15 **[Challenging]** • **Target: Derivatives of Inverse Functions** • **Cross-Lesson: 4.01 Rates of Change and Rectilinear Motion**

Find $\frac{d}{dx}[\sin((2x+1)^3)]$.

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16 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 4.02 Linear Approximation and L'Hôpital's Rule

Find $\frac{d}{dx}[\sin((2x-1)^3)]$.



17 [Challenging] • Target: First Derivative by Implicit Differentiation • Cross-Lesson: 5.01 Extrema and the Mean Value Theorem

Find $\frac{d}{dx}[\sin((2x-2)^3)]$.

18 [Challenging] • Target: Second Derivatives Implicitly • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find $\frac{d}{dx}[(\sin(-2x)+1)^3]$.

19 [Challenging] • Target: Derivatives of Inverse Functions • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find $\frac{d}{dx}[(\sin(-2x)+-1)^2]$.

20 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find $\frac{d}{dx}[(\sin(-2x)+-2)^2]$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: First Derivative by Implicit Differentiation

Find $\frac{dy}{dx}$ for the circle $x^2+y^2=25$.

a. $-\frac{x}{y}$

b. $\frac{x}{y}$

c. $-\frac{y}{x}$

d. $\frac{y}{x}$

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22 [Foundational] • Target: First Derivative by Implicit Differentiation

Find $\frac{dy}{dx}$ for the curve $x^3 + y^3 = 6xy$.

a. $\frac{2y - x^2}{y^2 - 2x}$

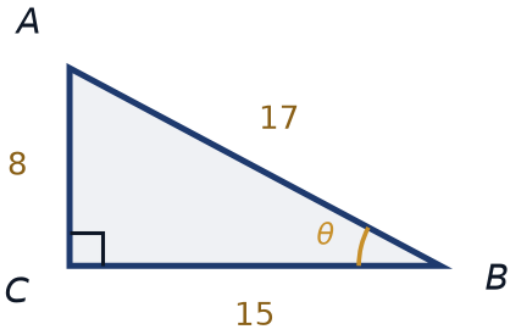
b. $\frac{x^2 - 2y}{2x - y^2}$

c. $\frac{2x - y^2}{x^2 - 2y}$

d. $\frac{x^2 + y^2}{6}$

23 [Foundational] • Target: First Derivative by Implicit Differentiation

Find the slope of the tangent line to $x^2y + y^2 = 5$ at the point $(2, 1)$.



a. $-\frac{2}{3}$

b. $-\frac{4}{5}$

c. $-\frac{4}{6}$

d. -1

24 [Foundational] • Target: Second Derivatives Implicitly

For $x^2 + y^2 = 16$, what is $\frac{d^2y}{dx^2}$ expressed solely in terms of y?

a. $-\frac{16}{y^3}$

b. $\frac{16}{y^3}$

c. $-\frac{1}{y^2}$

d. $-\frac{4}{y^3}$

25 [Foundational] • Target: Second Derivatives Implicitly

At the point $(0, 2)$ on the ellipse $4x^2 + y^2 = 4$, what is $\frac{d^2y}{dx^2}$?

a. -2

b. 2

c. -4

d. 0

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26 [Foundational] • Target: Second Derivatives Implicitly

What is the sign of the concavity at $(3, 4)$ on the circle $x^2 + y^2 = 25$?

a. Negative (concave down)

b. Positive (concave up)

c. Zero (inflection point)

d. Undefined

27 [Foundational] • Target: Derivatives of Inverse Functions

If g is the inverse of f, which formula gives $g'(a)$ where $f(b) = a$?

a. $g'(a) = \frac{1}{f'(b)}$

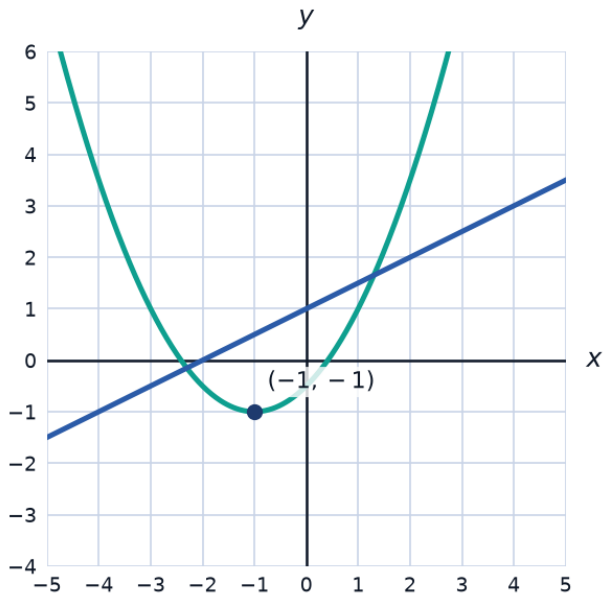
b. $g'(a) = f'(b)$

c. $g'(a) = \frac{1}{f'(a)}$

d. $g'(a) = -\frac{1}{f'(b)}$

28 [Foundational] • Target: Derivatives of Inverse Functions

Let $f(x) = x^3 + 2x - 1$. If g is the inverse of f, find $g'(2)$.



a. $\frac{1}{5}$

b. 5

c. $\frac{1}{14}$

d. 14

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Derivatives of Inverse Functions

Let $f(x) = x^5 + x + 1$. What is the derivative of f^{-1} at $x = 3$?

a. $\frac{1}{6}$

b. 6

c. $\frac{1}{80}$

d. $\frac{1}{5}$

30 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

What is $\frac{d}{dx}[\arcsin(3x)]$?

a. $\frac{3}{\sqrt{1-9x^2}}$

b. $\frac{1}{\sqrt{1-9x^2}}$

c. $\frac{3}{1+9x^2}$

d. $\frac{3}{\sqrt{1-3x^2}}$

31 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

Find $\frac{d}{dx}[\arctan(x^2)]$.

a. $\frac{2x}{1+x^4}$

b. $\frac{1}{1+x^4}$

c. $\frac{2x}{1+x^2}$

d. $\frac{2x}{\sqrt{1-x^4}}$

32 [Intermediate] • Target: Derivatives of Inverse Trigonometric Functions

What is the slope of the tangent line to $y = \arctan(x)$ at $x = 1$?

a. $\frac{1}{2}$

b. 1

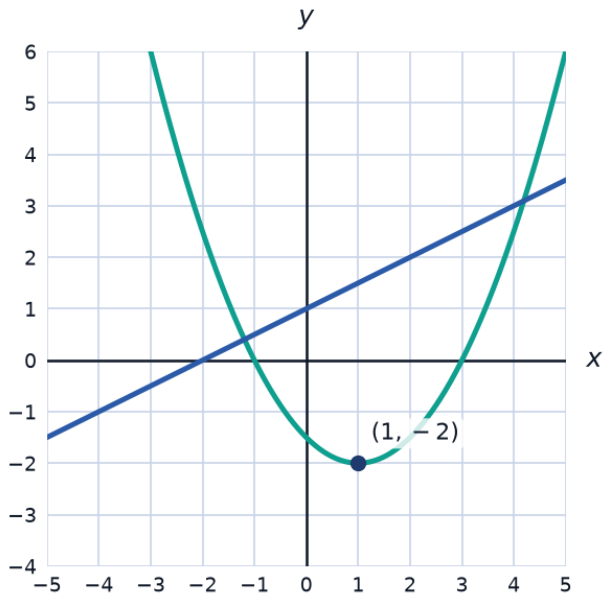
c. $\frac{\pi}{4}$

d. $\frac{1}{\sqrt{2}}$

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33 [Intermediate] • Target: First Derivative by Implicit Differentiation

Find $\frac{d}{dx}[(\sin(-2x)+2)^2]$.



a. $4-2\sin(2x)$

b. $-4(2-\sin(2x))\cos(2x)$

c. $4\sin(2x)-8$

d. $(4-2\sin(2x))\cos(2x)$

34 [Intermediate] • Target: Second Derivatives Implicitly

Find $\frac{d}{dx}[\sin((3x-2)^2)]$.

a. $(18x-12)\sin((3x-2)^2)$

b. $(18x-12)\cos((3x-2)^2)$

c. $(6x-4)\cos((3x-2)^2)$

d. $\cos((3x-2)^2)$

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35 **[Intermediate]** • **Target: Derivatives of Inverse Functions**

Find $\frac{d}{dx}[(\sin(-2x)+1)^2]$.

a. $4\sin(2x)-4$

b. $-4(1-\sin(2x))\cos(2x)$

c. $2-2\sin(2x)$

d. $(2-2\sin(2x))\cos(2x)$

36 **[Intermediate]** • **Target: Derivatives of Inverse Trigonometric Functions**

Find $\frac{d}{dx}[\sin((3x+2)^2)]$.

a. $\cos((3x+2)^2)$

b. $(6x+4)\cos((3x+2)^2)$

c. $(18x+12)\sin((3x+2)^2)$

d. $(18x+12)\cos((3x+2)^2)$

37 **[Intermediate]** • **Target: First Derivative by Implicit Differentiation**

Find $\frac{d}{dx}[\sin((1-2x)^2)]$.

a. $(8x-4)\sin((1-2x)^2)$

b. $(2-4x)\cos((1-2x)^2)$

c. $\cos((1-2x)^2)$

d. $(8x-4)\cos((1-2x)^2)$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Second Derivatives Implicitly • Cross-Lesson: 5.02 Concavity, Curve Sketching, and Optimization

Find $\frac{d}{dx}[\sin((3x+1)^2)]$.

a. $\cos((3x+1)^2)$

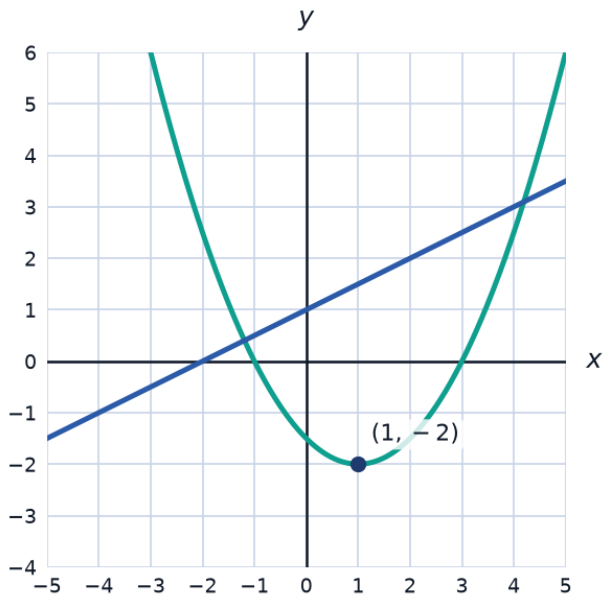
b. $(6x+2)\cos((3x+1)^2)$

c. $(18x+6)\cos((3x+1)^2)$

d. $(18x+6)\sin((3x+1)^2)$

39 [Challenging] • Target: Derivatives of Inverse Functions • Cross-Lesson: 6.01 Antiderivatives and Riemann Sums

Find $\frac{d}{dx}[\sin((3x-1)^2)]$.



a. $(18x-6)\sin((3x-1)^2)$

b. $(6x-2)\cos((3x-1)^2)$

c. $(18x-6)\cos((3x-1)^2)$

d. $\cos((3x-1)^2)$

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40 [Challenging] • Target: Derivatives of Inverse Trigonometric Functions • Cross-Lesson: 6.02 The Fundamental Theorem of Calculus

Find $\frac{d}{dx}[\sin((2-2x)^2)]$.

a. $(4-4x)\cos((2-2x)^2)$

b. $(8x-8)\sin((2-2x)^2)$

c. $\cos((2-2x)^2)$

d. $(8x-8)\cos((2-2x)^2)$

41 [Challenging] • Target: First Derivative by Implicit Differentiation • Cross-Lesson: 7.01 Differential Equations and Slope Fields

Find $\frac{d}{dx}[(\sin(3x)+\ -1)^3]$.

a. $3(\sin(3x)-1)^2$

b. $9(\sin(3x)-1)^2\cos(3x)$

c. $9(\sin(3x)-1)^2$

d. $3(\sin(3x)-1)^2\cos(3x)$

42 [Challenging] • Target: Second Derivatives Implicitly • Cross-Lesson: 8.01 Area Between Curves

Find $\frac{d}{dx}[(\sin(2x)+\ -1)^3]$.

a. $3(\sin(2x)-1)^2$

b. $6(\sin(2x)-1)^2\cos(2x)$

c. $6(\sin(2x)-1)^2$

d. $3(\sin(2x)-1)^2\cos(2x)$

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43 **[Challenging]** • **Target: Derivatives of Inverse Functions** • **Cross-Lesson: 8.02 Volumes by the Disc and Washer Methods**

Find $\frac{d}{dx}[\sin((2x-1)^2)]$.

a. $\cos((2x-1)^2)$

b. $(8x-4)\sin((2x-1)^2)$

c. $(8x-4)\cos((2x-1)^2)$

d. $(4x-2)\cos((2x-1)^2)$

44 **[Challenging]** • **Target: Derivatives of Inverse Trigonometric Functions** • **Cross-Lesson: 1.01 Limits Graphically and Numerically**

Find $\frac{d}{dx}[\sin((2x-2)^2)]$.

a. $(8x-8)\sin((2x-2)^2)$

b. $(8x-8)\cos((2x-2)^2)$

c. $(4x-4)\cos((2x-2)^2)$

d. $\cos((2x-2)^2)$

45 **[Challenging]** • **Target: First Derivative by Implicit Differentiation** • **Cross-Lesson: 1.02 Continuity and the Intermediate Value Theorem**

Find $\frac{d}{dx}[(\sin(3x) + \ -2)^3]$.

a. $3(\sin(3x)-2)^2\cos(3x)$

b. $9(\sin(3x)-2)^2$

c. $3(\sin(3x)-2)^2$

d. $9(\sin(3x)-2)^2\cos(3x)$

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